

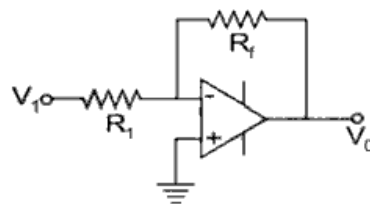
Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly in the answer sheet.

(Level/CO) Marks

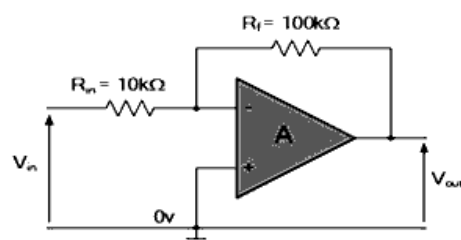
Q. 1 Solve any two of the followings.

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|--|---------|---|
| A) Explain the block diagram of op-amp? | Level 1 | 6 |
| B) Define: i) CMRR ii) Gain iii) Slew rate | Level 1 | 6 |
| C) If $V_{in}=2V$, $R_1= 200K\Omega$ and $R_f=500K\Omega$, find the output voltage of an Op-Amp. | Level 2 | 6 |



Q.2 Solve any two of the followings.

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|--|---------|---|
| A) Find the closed loop gain of the following inverting amplifier circuit. | Level 3 | 6 |
|--|---------|---|



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| B) Discuss the application of an Op-Amp as a differentiator? | Level 2 | 6 |
| C) Explain Clippers and Clampers using Op-Amp. | Level 1 | 6 |

Q. 3 Solve any two of the followings.

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| A) With suitable diagrams, explain a summing amplifier using Op-amp. | Level 1 | 6 |
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| B) Explain the concept of a non-inverting amplifier and virtual ground in Op-Amp. | Level 1 | 6 |
| C) Design a circuit using Op-amp that provides a gain of 5.5. Assume $R_{i=}$ 10k Ω , what value would you choose for R_f ? | Level 2 | 6 |

Q.4 Solve any two of the followings.

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|--|----------------|----------|
| A) $R = 10\text{ k}$, $v_1 = 2.011\text{V}$, and $v_2 = 2.017\text{V}$. If R_G is adjusted to 500 , determine: (a) The voltage gain, (b) The output voltage v_o . | Level 1 | 6 |
| B) Define active filter and discuss its classification? | Level 1 | 6 |
| C) Explain the construction and working of a relaxation oscillator. | Level 1 | 6 |

Q. 5 Solve any two of the followings.

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|--|----------------|----------|
| A) Compare RC oscillators with LC oscillators. | Level 1 | 6 |
| B) Explain the working of Hartley oscillator with neat diagram. | Level 2 | 6 |
| C) Explain the concept of a weighted resistor. List its advantages, disadvantages. | Level 1 | 6 |

***** End *****